

Modular and Auditable Clinical Extraction from Epilepsy Letters

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Abstract—Clinical extraction systems need more than a single aggregate score: their model, deterministic rules, repairs, evidence, scorer, and evaluation boundary must remain separately inspectable. We evaluate a modular architecture on two epilepsy-letter tasks: current seizure-frequency labeling on the Gan 2026 synthetic benchmark and broad epilepsy phenotyping on ExECTv2. For each task we retain rules-only, LLM-only, and hybrid reference cells with executable source closure and no-call replay. On the frozen Gan holdout, the operational single-pass structured-event system reaches 364/450 Purist accuracy, while a more complex multi-trace ceiling reaches 379/450. On ExECT dev140, the retained deterministic, LLM-only, and hybrid references reach 0.3548 strict per-item F1, 0.7393 clinical-headline F1, and 0.9189 clinical-headline F1, respectively. A bounded same-core full-200 aggregate comparison spans 0.8197–0.8566 clinical-headline F1 across Qwen 3.6 35B, GPT-4.1-mini, and DeepSeek chat. Saved-output ablations show positive normalization contributions on both tasks, while the evidence check is score-inert on those representative replays. The contribution is a reproducible, stage-owned comparison with explicit claim boundaries, not a claim of independent clinical validation, strict ExECT benchmark reproduction, or a completed six-model result.

Index Terms—clinical information extraction, epilepsy, large language models, hybrid systems, reproducibility, evidence provenance

I. INTRODUCTION

Epilepsy letters combine temporal reasoning, clinical terminology, medication regimens, investigation findings, and ambiguous current-versus-historical statements. Large language models (LLMs) can broaden extraction, but a model call can hide where a result came from. Deterministic systems are easier to inspect, yet often fail on linguistic variation and long-range clinical selection.

We study these approaches as explicit architecture families rather than as one opaque pipeline. The same project contains a deep single-label task and a broad multi-entity task. This makes it possible to ask which stages transfer, which results depend on task-specific scoring, and where evidence is insufficient for a stronger claim.

The paper contributes: (1) a retained two-task by three-family comparison with exact replay closure; (2) explicit ownership of extraction, normalization, projection, repair, evidence validation, and scoring; (3) frozen aggregate evidence for the Gan operational-versus-ceiling trade-off and a three-model ExECT transfer study; and (4) reproducibility controls that pin prompts, scorers, splits, repair policies, runtime identifiers, dependencies, and artifact hashes.

II. RELATED WORK

ExECT demonstrated structured epilepsy extraction from unstructured clinic letters and later work documented the corresponding annotation resource [1], [2]. Seizure-frequency extraction has also been studied with machine-reading and pretrained-language-model approaches [3], [4], while recent work has explored fine-tuned LLMs and reproducible synthetic clinical letters [5], [6]. Our focus is complementary: pipeline stages and evidence boundaries are part of the scientific object, and rules-only, LLM-only, and hybrid forms are compared without collapsing their different scoring and repair mechanisms.

III. METHODS

A. Tasks and Split Policy

Gan 2026 asks for one current seizure-frequency label per synthetic letter [6]. Validation750 is the row-inspectable development and replay surface. Test450 is a locked holdout: only frozen aggregate results are retained, and row-level holdout failures are not used for development.

ExECTv2 covers Diagnosis, SeizureFrequency, Prescription, and Investigations in the primary comparison, with additional entities retained in the deterministic all-nine reference [1], [2]. Dev140 is the row-inspectable development surface. Full200 combines dev140 with held-out test60 and is reported only as a development-inclusive aggregate audit; it is not an independent holdout.

B. Architecture Families

For each task, the retained manifest contains exactly one rules-only, LLM-only, and hybrid cell. Rules-only cells use deterministic extraction, normalization, projection, validation, and rendering without model calls. LLM-only cells concentrate clinical extraction in one model program. The ExECT GEPA cell is a negative development comparator, not the production control. Hybrid cells keep model-owned clinical extraction distinct from deterministic selection, normalization, evidence checks, and scoring. Gan uses one structured-event pass; ExECT uses manifest-driven finding assembly with family-specific lenses.

C. Scoring, Repair, and Attribution

Gan uses Purist and Pragmatic label scoring; Purist is the primary strict label surface. ExECT’s primary research-control surface is de-duplicated `clinical_headline` recovery across four families. Phrase, CUI, evidence-valid, and full-attribute companions are reported separately.

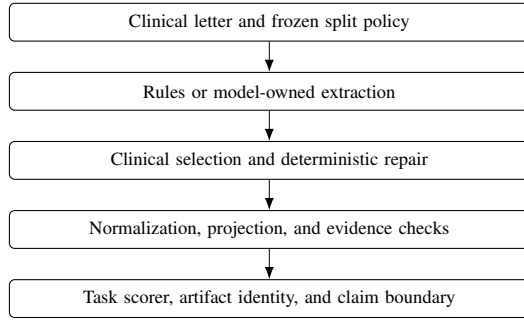


Fig. 1. Stage-owned extraction spine. Ownership remains explicit even when a stage is implemented by an LLM.

TABLE I
RETAINED TWO-TASK BY THREE-FAMILY REFERENCE CELLS

Task	Family	Split	Retained result	Role and boundary
ExECTv2	Rules only	dev140	Strict item F1 0.3548	Incomplete deterministic development reference.
ExECTv2	LLM only	dev140	Headline F1 0.7393	GEPA negative development comparator.
ExECTv2	Hybrid	dev140	Headline F1 0.9189	Current development performance control.
Gan 2026	Rules only	validation750	697/750 Purist	Deterministic development comparator.
Gan 2026	LLM only	validation750	581/750 Purist	Single-pass development comparator.
Gan 2026	Hybrid	validation750	661/748 rendered Purist	Single-pass structured-event comparator.

`clinical_headline` is not presented as reproduction of the published strict ExECT benchmark.

Raw model selection, JSON or format repair, source-exact evidence repair, deterministic semantic repair, projection, and final scoring are separate stages. Model-specific transport or output-shape adapters do not become model quality evidence. Semantic repairs retain a named deterministic owner and must be ablated or otherwise tested before supporting a causal interpretation.

D. Architecture Freeze

The 14 July 2026 reduced-reference-architecture freeze pins the implementation at commit 46562134. Manifest v3 fingerprints the dependency declaration and lock, prompt contracts, scorers, split manifests, repair policies, model policy, split runbook, and CI workflow. A semantic prompt, scorer, split, repair, runtime-route, or component-graph change requires a new freeze and complete replay. The freeze does not itself authorize model calls.

IV. RESULTS

A. Retained Architecture Comparison

Table I is the minimum retained scientific comparison. Scores are not compared across tasks because the datasets, outputs, and scorers differ.

All six cells replay from current code and retained outputs without model calls. The ExECT hybrid replay also returns evidence-valid F1 0.8913. Its current benchmark/CUI companion is 0.4791 versus 0.4729 in the retained run artifact;

TABLE II
GAN OPERATIONAL AND CEILING EVIDENCE ON LOCKED TEST450

System	Purist	Role
Single structured-event pass	364/450	Operational close-off
V12 multi-trace reviewer	379/450	Complexity ceiling

TABLE III
EXECT SAME-CORE FULL200 AGGREGATE `CLINICAL_HEADLINE`

Model condition	Overall	Diagnosis	SF	Prescription	Investigations	Call/parse failures
DeepSeek chat	0.8566	0.8708	0.7602	0.8926	0.9091	0/1
GPT-4.1-mini	0.8356	0.8397	0.7525	0.8926	0.8563	0/0
Qwen 3.6 35B repair v02	0.8197	0.8307	0.7020	0.8926	0.8503	0/0

this companion-scorer drift is open and does not affect the reproduced 0.9189 headline result.

B. Gan Frozen Holdout Evidence

Both rows in Table II are frozen aggregate evidence. The 15-row quality difference does not yet support an efficiency conclusion. A matched table of calls, tokens, cost, latency, hardware, and cache policy remains open.

C. ExECT Same-Core Model Evidence

The retained model-transfer package holds the two-call no-SF-adjudicator component graph fixed while changing the runtime model. Table III is a full200 development-inclusive aggregate read; no test60 row-level analysis is used.

These runs support a bounded same-core transfer statement. They are not a strict same-prompt panel: the retained GPT condition used temperature 0.3, and Qwen used a compact prompt and output-contract repair. The requested six-model comparison remains incomplete, with three missing runtime conditions requiring predeclaration.

D. Cross-Task Components and Reliability

The zero score delta in Table IV does not show that evidence validation is unnecessary. A filter may be essential on malformed or hallucinated outputs even when all selected reference predictions already satisfy it. Rejection and repair tests, not aggregate F1 alone, carry that part of the evidence.

Model-reported confidence remains unused, and no low-burden review-routing policy is promoted. The retained manifest does not include evidence for a cross-task Wall-transfer claim.

V. DISCUSSION

The strongest result is structural rather than universal. The reduced system can reproduce six architecture cells while

TABLE IV
SAVED-OUTPUT COMPONENT ABLATION

Component	ExECT dev140	Gan validation750
Normalization/dictionary	+0.0389	+0.0293
Evidence validation	0.0000	0.0000

TABLE V
RETAINED RELIABILITY EVIDENCE AND LIMITS

Subject	Retained result	Boundary
ExECT evidence	Hybrid dev140 minimum exact evidence rate 1.0000	Development reference.
ExECT calibration	Brier 0.2225; base-rate Brier 0.2340; ECE 0.0587	Full200 aggregate internal scoring-rule result.
Gan grounding	Architecture-specific validation grounding packages retained	Metrics are not uniform across families.
Reproducibility	1,153 tests, Ruff, mypy, hashes, split barriers, and six-cell replay pass on Python 3.11	Engineering verification, not clinical validation.

keeping component ownership and evaluation boundaries explicit. Hybrid assembly is substantially stronger than the retained ExECT rules-only and LLM-only development references on `clinical_headline`, but the three cells do not share a strict published benchmark surface. Deterministic paper-comparable engineering remains open.

The model-transfer result shows that the component graph runs with three qualitatively different runtime models. DeepSeek leads the retained full200 aggregate, while Qwen trails most clearly on SeizureFrequency. Because runtime conditions are asymmetric and the panel is only three of six, this is evidence of bounded portability rather than a universal model-agnostic claim.

The cross-task ablation suggests that normalization is useful in both tasks. The evidence gate’s zero F1 delta illustrates why clinical pipeline evidence cannot be reduced to aggregate score changes: rejection, provenance, and repair behavior need direct contract tests.

VI. LIMITATIONS

Gan test450 is frozen aggregate evidence and cannot support row-level tuning. ExECT full200 includes dev140 and is not an independent holdout. `clinical_headline` is a clinical-recovery surface, not the published strict benchmark. The ExECT benchmark/CUI companion has small current-code replay drift. The model panel is three of six and uses asymmetric runtime conditions. Model-reported confidence and low-burden review routing are not validated. Annotation-quality findings are internally adjudicated; unqualified clinical validity would require independent domain review. The current retained evidence does not support an ExECT Wall-transfer claim.

VII. CONCLUSION

A small, frozen, stage-owned clinical extraction system can preserve both performance evidence and the provenance needed to interpret it. The retained results support a two-task rules/LLM/hybrid comparison, a bounded Gan operational-versus-ceiling result, and a three-model ExECT same-core study. They do not yet support strict ExECT benchmark reproduction, a six-model conclusion, or independent clinical validation. Those boundaries are part of the result, not footnotes to it.

REPRODUCIBILITY STATEMENT

The machine-readable retained-evidence manifest records the six reference cells, selected artifacts and hashes, policy fingerprints, and no-call replay expectations. The synchronized Markdown manuscript and this IEEE source use only evidence retained by that manifest and claim strength permitted by the paper-provenance canon.

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